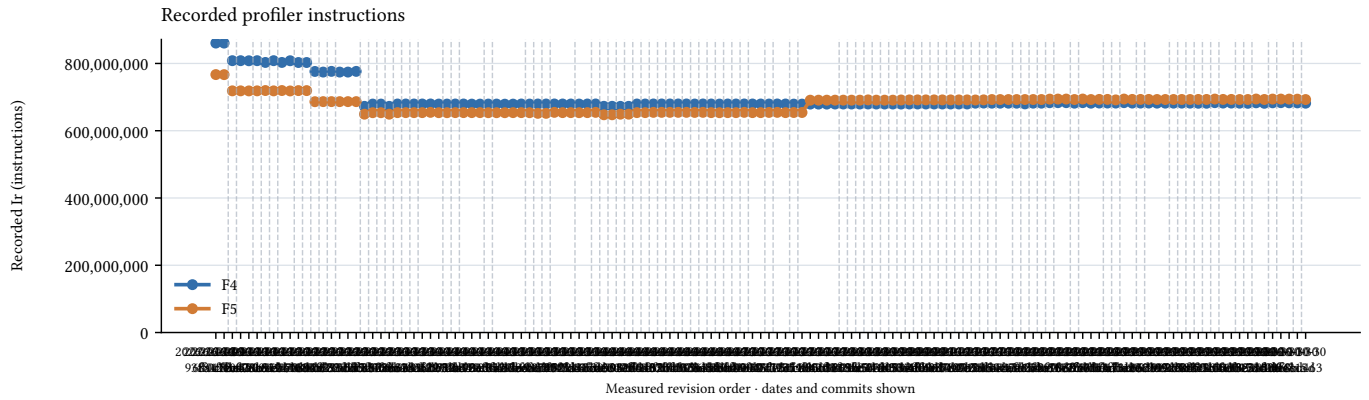
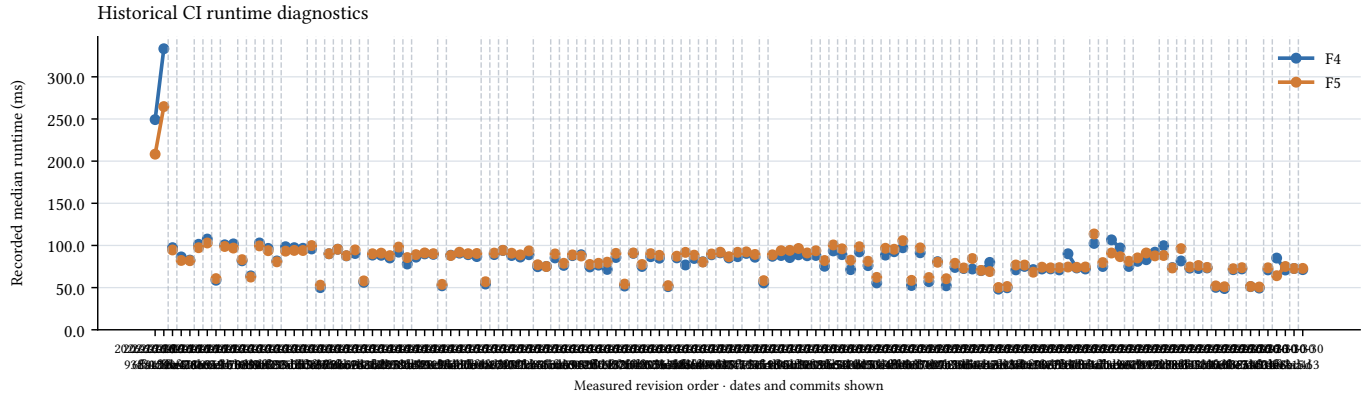
# Boolean matrix step: F4 and F5

One shared K_1 / 2^23 m=2 system, 22 Boolean variables, maximum degree 4. Matrix construction, reduction and readback; F5 includes its criterion.




133 archived revisions; recorded historical profiler scope, worker coverage incomplete.
Gaps mark changed outputs or measurement conditions. Sources: plot-data.json.




133 archived revisions; recorded medians, no stored spread or isolated timing eligibility.
CI runtime diagnostics; sample counts and condition changes retained in plot-data.json.

| UTC date / revision | F4 Ir | F5 Ir | F4 ms | F5 ms |
|---|---|---|---|---|
| 2026-10-08 / 93ff0c72 | 860,633,889 | 766,940,469 | 249.241 | 208.250 |
| 2026-10-09 / 6f81260f | 860,555,567 | 766,776,087 | 333.279 | 264.687 |
| 2026-10-11 / 8a41eca7 | 808,263,705 | 718,576,235 | 97.380 | 94.658 |
| 2026-10-11 / 0cfba411 | 808,263,705 | 718,578,523 | 86.390 | 82.214 |
| 2026-10-11 / f88e5909 | 807,989,935 | 718,391,055 | 82.597 | 81.948 |
| 2026-10-11 / fb926c02 | 808,263,705 | 718,576,189 | 101.379 | 97.323 |
| 2026-10-11 / e478b952 | 803,017,657 | 719,516,063 | 107.585 | 102.927 |
| 2026-10-11 / aa0cee7b | 808,281,869 | 718,492,761 | 58.723 | 60.673 |
| 2026-10-12 / 81ca1c78 | 803,015,877 | 719,509,003 | 100.762 | 98.775 |
| 2026-10-12 / a8194d61 | 808,072,539 | 718,282,030 | 101.827 | 96.904 |
| 2026-10-12 / 5cbe16d5 | 802,824,065 | 719,316,911 | 81.784 | 83.144 |
| 2026-10-12 / eb70862e | 802,824,065 | 719,316,876 | 63.944 | 62.505 |
| 2026-10-12 / 31088bfc | 776,242,363 | 686,315,894 | 102.929 | 99.489 |
| 2026-10-12 / 0d34f782 | 774,379,185 | 686,521,131 | 96.521 | 93.790 |
| 2026-10-12 / 89e3233c | 776,490,367 | 686,543,371 | 81.628 | 80.581 |
| 2026-10-12 / 80f5391d | 774,379,185 | 686,519,896 | 98.501 | 93.239 |
| 2026-10-12 / c62555b7 | 774,379,185 | 686,519,896 | 97.360 | 94.117 |
| 2026-10-12 / 633d2158 | 776,467,358 | 686,512,000 | 96.637 | 94.008 |
| 2026-10-12 / ca0d5f86 | 672,500,776 | 649,174,334 | 95.848 | 99.895 |
| 2026-10-12 / b5cc6ffc | 679,242,389 | 653,622,849 | 49.965 | 52.882 |
| 2026-10-13 / bf92906b | 679,242,389 | 653,622,849 | 90.163 | 90.201 |
| 2026-10-13 / d85756aa | 672,486,486 | 649,160,075 | 95.497 | 95.480 |
| 2026-10-13 / ab28b57b | 679,449,939 | 653,823,121 | 87.697 | 87.878 |
| 2026-10-13 / 56f66e43 | 679,449,939 | 653,828,331 | 90.448 | 94.715 |
| 2026-10-13 / 28b1daf2 | 679,449,939 | 653,828,331 | 56.215 | 57.961 |
| 2026-10-14 / cc962159 | 679,449,939 | 653,831,656 | 88.546 | 90.149 |
| 2026-10-14 / 33816db5 | 679,549,997 | 654,627,509 | 88.344 | 90.874 |
| 2026-10-14 / 191a0739 | 679,240,609 | 653,664,759 | 85.135 | 87.917 |
| 2026-10-14 / cd226d0c | 679,240,609 | 653,664,759 | 92.059 | 98.073 |
| 2026-10-14 / b2272a44 | 679,449,939 | 653,873,401 | 78.007 | 85.621 |
| 2026-10-14 / bc82d128 | 679,449,939 | 653,873,401 | 86.126 | 89.102 |
| 2026-10-14 / 912a98ed | 679,098,623 | 653,830,968 | 89.995 | 91.125 |
| 2026-10-14 / 8d8b1295 | 679,098,623 | 653,830,968 | 89.395 | 90.230 |

| UTC date / revision | F4 Ir | F5 Ir | F4 ms | F5 ms |
|---|---|---|---|---|
| 2026-10-14 / b339e847 | 679,240,608 | 653,621,583 | 52.150 | 53.839 |
| 2026-10-15 / 7912faa5 | 679,226,318 | 653,606,027 | 88.406 | 88.091 |
| 2026-10-15 / 3c5d435e | 678,905,208 | 653,631,022 | 90.855 | 91.956 |
| 2026-10-15 / f0599bf9 | 679,098,622 | 653,825,706 | 89.060 | 90.253 |
| 2026-10-15 / 3db0ecbc | 679,449,938 | 653,830,913 | 86.813 | 90.492 |
| 2026-10-15 / addb6a02 | 679,256,525 | 653,632,330 | 54.248 | 56.951 |
| 2026-10-15 / 12348e50 | 679,449,939 | 651,714,930 | 89.150 | 90.941 |
| 2026-10-15 / b906df02 | 679,449,939 | 651,714,179 | 94.066 | 94.227 |
| 2026-10-15 / cf274521 | 679,549,984 | 654,627,503 | 88.061 | 90.795 |
| 2026-10-15 / 1b874fcc | 679,449,939 | 653,830,975 | 86.246 | 88.831 |
| 2026-10-15 / 700f7d31 | 679,444,579 | 653,824,679 | 89.023 | 93.545 |
| 2026-10-15 / 289bb5fb | 679,264,149 | 653,687,604 | 74.802 | 77.102 |
| 2026-10-16 / 92246530 | 679,449,938 | 653,829,966 | 75.128 | 75.009 |
| 2026-10-16 / 3a4a5cec | 679,549,995 | 654,621,347 | 85.229 | 89.944 |
| 2026-10-16 / 333c08c0 | 672,605,656 | 647,427,799 | 76.341 | 78.831 |
| 2026-10-16 / 4cfcb94e | 672,605,656 | 647,427,799 | 87.944 | 88.781 |
| 2026-10-16 / ca0310ff | 672,486,486 | 649,202,105 | 89.012 | 87.501 |
| 2026-10-16 / de0c9391 | 672,500,776 | 649,216,408 | 74.498 | 77.487 |
| 2026-10-16 / 7051f4e2 | 679,449,938 | 653,873,407 | 76.675 | 78.834 |
| 2026-10-16 / 082d5507 | 679,445,034 | 653,868,507 | 71.580 | 80.148 |
| 2026-10-16 / 2f02e32c | 679,549,984 | 654,627,503 | 85.711 | 90.622 |
| 2026-10-16 / 6d437f0c | 679,549,984 | 654,627,503 | 51.852 | 54.190 |
| 2026-10-16 / 6bf2bcf0 | 679,549,997 | 654,621,861 | 90.619 | 91.074 |
| 2026-10-16 / f8300065 | 679,549,997 | 654,669,949 | 75.064 | 77.431 |
| 2026-10-16 / ccf43056 | 679,549,997 | 654,669,949 | 86.686 | 90.254 |
| 2026-10-16 / 57b028d8 | 679,350,531 | 654,421,882 | 84.863 | 88.297 |
| 2026-10-16 / f25d5e52 | 679,549,984 | 654,669,936 | 51.138 | 52.369 |
| 2026-10-16 / 213e7a80 | 679,449,939 | 653,830,971 | 85.272 | 87.253 |
| 2026-10-16 / 78adfda4 | 679,240,609 | 653,621,067 | 76.846 | 91.802 |
| 2026-10-17 / b1bf5b89 | 679,449,939 | 653,825,765 | 84.571 | 88.476 |
| 2026-10-17 / 40bcf690 | 679,449,939 | 653,825,765 | 80.639 | 80.305 |
| 2026-10-17 / 8efd4e48 | 679,549,984 | 654,622,293 | 88.930 | 89.875 |
| 2026-10-17 / 89036527 | 679,449,939 | 653,830,971 | 91.230 | 91.994 |
| 2026-10-17 / a90646f1 | 679,435,649 | 653,817,360 | 85.170 | 86.662 |
| 2026-10-17 / b59187e5 | 679,340,625 | 654,418,146 | 86.682 | 91.890 |
| 2026-10-17 / fd043d4f | 679,549,955 | 654,621,012 | 90.548 | 92.367 |
| 2026-10-17 / 17c71374 | 679,449,939 | 653,830,397 | 86.043 | 89.344 |
| 2026-10-17 / 7d751c8d | 679,098,491 | 654,182,814 | 55.523 | 58.269 |
| 2026-10-17 / a95fcd95 | 679,098,493 | 654,182,133 | 87.157 | 88.860 |
| 2026-10-17 / 2c1164df | 679,198,639 | 690,798,036 | 88.066 | 93.755 |
| 2026-10-17 / e59342d3 | 679,198,639 | 690,798,036 | 85.604 | 94.177 |
| 2026-10-17 / ca00c151 | 679,198,639 | 690,798,036 | 88.918 | 96.426 |
| 2026-10-18 / f8501268 | 679,449,938 | 690,826,836 | 87.801 | 91.097 |
| 2026-10-18 / a1d0f1e6 | 679,449,938 | 690,826,836 | 88.118 | 93.700 |
| 2026-10-18 / c8613fe5 | 679,449,938 | 690,825,106 | 75.312 | 82.144 |
| 2026-10-18 / 3c192c34 | 679,449,938 | 690,825,106 | 93.514 | 100.589 |
| 2026-10-18 / 805b6279 | 679,549,993 | 691,623,334 | 89.125 | 96.021 |
| 2026-10-19 / 21d553b1 | 679,453,166 | 690,824,740 | 71.297 | 82.510 |
| 2026-10-19 / fc5d4835 | 679,449,937 | 690,826,048 | 92.362 | 98.482 |
| 2026-10-20 / 841a45cd | 679,451,718 | 690,805,442 | 76.066 | 81.266 |
| 2026-10-20 / 0e3c51ed | 679,340,663 | 691,396,123 | 55.582 | 62.124 |
| 2026-10-20 / d0337ff5 | 679,551,773 | 691,602,405 | 88.546 | 96.658 |
| 2026-10-20 / d30a63b8 | 679,549,994 | 691,600,618 | 92.571 | 95.491 |
| 2026-10-20 / 892ef408 | 679,549,994 | 691,600,618 | 97.275 | 105.731 |
| 2026-10-21 / b4f05610 | 679,551,773 | 691,602,032 | 52.495 | 58.611 |
| 2026-10-21 / 43e01b10 | 679,551,773 | 691,602,032 | 91.335 | 97.133 |
| 2026-10-22 / 02b37807 | 679,551,773 | 691,602,032 | 57.057 | 61.964 |
| 2026-10-23 / 6ed792e8 | 679,549,983 | 691,604,220 | 80.788 | 80.079 |
| 2026-10-23 / d34b9b39 | 679,549,983 | 691,604,220 | 52.168 | 60.548 |
| 2026-10-27 / 7e271266 | 681,943,949 | 691,347,680 | 73.901 | 78.829 |
| 2026-10-27 / c402a154 | 682,282,779 | 691,707,737 | 72.619 | 73.294 |
| 2026-10-27 / 1c756bcb | 681,899,901 | 692,697,266 | 72.047 | 84.493 |
| 2026-10-28 / a5db1413 | 681,899,901 | 692,697,266 | 70.151 | 70.376 |
| 2026-10-28 / 0650ca8b | 681,899,901 | 692,697,266 | 79.912 | 69.027 |
| 2026-10-28 / d4e85a6a | 681,885,611 | 692,687,520 | 48.174 | 50.139 |
| 2026-10-28 / e74711b4 | 679,810,476 | 692,710,161 | 49.796 | 51.316 |
| 2026-10-28 / cafd2389 | 681,943,647 | 692,740,867 | 71.045 | 76.833 |
| 2026-10-28 / eec5dcc3 | 681,943,647 | 692,740,867 | 75.109 | 76.743 |

| UTC date / revision | F4 Ir | F5 Ir | F4 ms | F5 ms |
|---|---|---|---|---|
| 2026-10-28 / 98a2f7fa | 683,245,465 | 693,927,563 | 71.387 | 68.293 |
| 2026-10-28 / 3f0e13ee | 683,415,446 | 694,093,622 | 72.074 | 74.215 |
| 2026-10-28 / 7be0960b | 683,415,446 | 694,093,622 | 72.232 | 73.439 |
| 2026-10-28 / a988ff46 | 681,943,879 | 692,741,225 | 70.987 | 73.705 |
| 2026-10-28 / 8177cd08 | 683,437,564 | 694,188,206 | 90.137 | 74.414 |
| 2026-10-28 / 0d839d5b | 681,899,901 | 692,698,574 | 73.298 | 74.059 |
| 2026-10-28 / 7fe71c4a | 682,335,928 | 693,119,282 | 72.101 | 74.442 |
| 2026-10-29 / 25d0e671 | 682,335,928 | 693,119,282 | 102.392 | 113.613 |
| 2026-10-29 / 0fc7cdad | 682,218,335 | 691,617,426 | 75.044 | 79.697 |
| 2026-10-29 / 3f1dcd9e | 683,439,590 | 694,127,041 | 106.645 | 91.019 |
| 2026-10-29 / 0af11c62 | 682,263,636 | 693,066,268 | 97.025 | 86.703 |
| 2026-10-29 / f73e667b | 682,263,636 | 693,066,268 | 75.045 | 81.275 |
| 2026-10-29 / be2969ce | 682,263,636 | 693,066,268 | 81.162 | 85.238 |
| 2026-10-29 / ca0c809d | 682,037,440 | 692,830,863 | 83.099 | 91.075 |
| 2026-10-29 / 85d80508 | 682,037,440 | 692,830,863 | 92.024 | 87.672 |
| 2026-10-29 / 91902d0c | 682,037,440 | 692,830,863 | 99.730 | 88.072 |
| 2026-10-29 / 117c9c4f | 682,076,130 | 692,882,707 | 73.768 | 73.314 |
| 2026-10-29 / 992c2579 | 682,032,151 | 692,835,720 | 81.606 | 96.247 |
| 2026-10-29 / 2c583c5c | 681,686,676 | 692,985,843 | 73.089 | 74.436 |
| 2026-10-29 / 78f84234 | 681,686,676 | 692,980,550 | 72.621 | 76.212 |
| 2026-10-29 / 8eb04d5e | 683,400,900 | 694,082,886 | 73.332 | 73.948 |
| 2026-10-29 / 055c5035 | 682,218,705 | 693,065,470 | 50.283 | 52.017 |
| 2026-10-29 / 11b1130e | 682,218,705 | 693,065,470 | 48.863 | 51.023 |
| 2026-10-29 / 21f66351 | 681,905,560 | 692,750,765 | 71.271 | 72.453 |
| 2026-10-30 / 00d005d4 | 681,510,817 | 692,810,040 | 72.133 | 73.652 |
| 2026-10-30 / a0be31d0 | 683,020,225 | 694,203,117 | 51.174 | 51.359 |
| 2026-10-30 / f32aa536 | 681,568,096 | 692,878,781 | 49.454 | 50.717 |
| 2026-10-30 / 178c03f4 | 683,014,384 | 694,042,678 | 70.999 | 73.487 |
| 2026-10-30 / 50f24e8b | 683,419,396 | 694,100,165 | 85.036 | 64.112 |
| 2026-10-30 / 01df67aa | 682,999,367 | 694,032,870 | 70.858 | 75.058 |
| 2026-10-30 / 1d9e815d | 682,791,459 | 693,824,962 | 72.658 | 72.715 |
| 2026-10-30 / af6ed463 | 681,856,292 | 692,705,445 | 71.310 | 72.802 |

Historical Python-orchestrated perf.history/v1 snapshots; no new performance run or speedup claim.

Runtime: one recorded median per kernel and revision; sample counts retained per point, no stored spread, no isolated timing eligibility. Runtime is diagnostic.

Profiler Ir is the recorded historical scope. Worker-thread coverage is incomplete for these kernels; it is not a complete solver operation count.

Lines stop at missing values, changed output fingerprints, or changed host/compiler/harness/profiler. Different algorithm fingerprints are expected and are not equated.

X positions are measured revision order, labelled with UTC commit date and revision; spacing is not elapsed time.

Evidence: 133 imported perf.history/v1 snapshots; historical_sources in plot-data.json retains raw filenames, SHA-256 hashes, hosts, harnesses and sample counts. Source data and provenance: plot-data.json and REPORT.md. These plots do not establish an end-to-end ECDLP gain.